CS & IT ENGINEERING

Algorithms

Analysis of Algorithms

Lecture No.- 07

By- Aditya sir

Recap of Previous Lecture

Topic

Topic

Properties

Imp Practice Questions.

Topics to be Covered

Topic

Topic

Topic

Analysis of an Algo.

Practice Questions

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

@ADITYASIR_PW

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Discrete Properties of Asymptotic Notations:

	Reflexive	Symmetric	Transitive	Transpose Symmetry
O	✓	×	✓	✓
Ω	✓	×	✓	✓
θ	✓	✓	✓	×
o	×	×	✓	✓
ω	×	×	✓	✓

~~[MCQ]~~ → MSQ

$\boxed{A, B, C}$ ✓

PYQ

#Q. Which of the following is TRUE.

given
1. f(n) is O(g(n))
2. g(n) is NOT O(f(n))
3. g(n) is O(h(n))
4. h(n) is O(g(n))

AJ Sir Conclusion Analysis Mtd

$\boxed{f < g = h}$

A. f(n) is O(h(n)) $\Rightarrow \boxed{f \le h}$ ✓

B. h(n) ≠O(f(n)) ✓

$h \nleq f \Rightarrow \boxed{h > f}$

C. (f(n) + h(n)) is O(g(n) + h(n)) ✓

$f + \cancel{h} \le g + \cancel{h}$

D. (f(n).g(n)) ≠ O(g(n). h(n))

$f \times \cancel{g} \nleq \cancel{g} \times h$

$f \nleq h \Rightarrow \boxed{f > h}$

given:

1) $f(n) = O(g(n)) \Rightarrow f \leq g$

2) $g(n) \neq O(f(n)) \Rightarrow g \nleq f \Rightarrow g > f$

$\boxed{f < g}$

3) $g(n) = O(h(n)) \Rightarrow g \leq h$

4) $h(n) = O(g(n)) \Rightarrow h \leq g$

$\boxed{g = h}$

Conclusion $\longrightarrow$ $\boxed{f < g = h}$

~~[MCQ]~~ MSQ

#Q. $f(n) = 2^n; g(n) = n^n$

Conclusion: $f < g \Rightarrow \boxed{f = O(g)}$

$f < g$

$f \geq g$ ✗

A $f(n) = O(g(n))$ $f \leq g$

C $f(n) = \Omega(g(n))$

B $f(n) = \theta(g(n))$

D None of these

$f = g$ ✗

Soln: $f(n) = 2^n$

$g(n) = n^n$

$2^n \quad n^n$

$2 < n$

$2^n < n^n$

$f < g$

$f = O(g) \Leftarrow f < g$

$2 < 3$

$2 \leq 3$ ✓

$2^n \quad n^n$

$\log(2^n) \quad \log(n^n)$

$n \quad n\log n$

$1 < \log n$

Topic : Analysis of Algorithms

(1) Every small oh is also Big Oh, but every Big Oh may or may not be small oh.

(2) Every small omega is also Big Omega, but every Big Omega may or may not be small omega.

$$O \quad \leq \quad <$$

$$\Omega \quad \geq \quad >$$

[MSQ]

#Q. $f(n) = n.2^n; g(n) = 4^n$

$\boxed{f < g}$

$f \leq g$

A $f(n) = O(g(n))$ $\longrightarrow$ 58.3%

C $f(n) = \Omega(g(n))$

$f \geq g$ ✗

B $f(n) = \theta(g(n))$

$f = g$ ✗

D None of these ✗

$f(n) = n \times 2^n$

$g(n) = 4^n \rightarrow 2^n \times 2^n$

$n \times \cancel{2^n} \qquad 2^n \times \cancel{2^n}$

$4^n = (2^2)^n = 2^{2n} = 2^{n+n}$

$n < 2^n$

$n \times 2^n < 4^n$

$f < g$

[MCQ]

PYQ

#Q. Let w(n) and A(n) represent respectively, the worst case and average case running time of an algorithm with input size of n, Which is always TRUE?

$\boxed{B \le A \le W}$

64%

D

$A < W$ → sometimes

A $A(n) = o(w(n))$ ✗

$A = W$

B $A(n) = \theta(w(n))$ ✗

$A > W$ → never True

E $A(n) = \omega(w(n))$ ✗

$A \geq W$ → sometimes True

C $A(n) = \Omega(w(n))$ ✗

$A \le W$ → always True

D $A(n) = O(w(n))$ ✓

[MSQ]

#Q. Asymptotic Comparison of 2 functions:

$f(n) = n$

$g(n) = n \log n$

$n < n\log n$

$f = o(g)$

$\cancel{n} < \cancel{n}\log n$

$1 < \log n$

A $f = O(g)$ ✓

B $f = o(g)$ ✓

C $f = \Omega(g)$ ✗

D $f = \omega(g)$ ✗

NAT

2m

0.51

0

0.51

$\boxed{0.5}$

110

11TBX

60 → 11TB

40 → 11TB

[MSQ]

#Q. Asymptotic Comparison of 2 functions:

$f(n) = n^2 (\log n)$

$g(n) = n (\log n)^{10}$

$\boxed{f > g}$

$f = \omega(g)$

A) $f = O(g)$ ✗

B) $f = \Omega(g)$ ✓

C) $f = \omega(g)$ ✓

D) $f = o(g)$ ✗

$$f = n^2 \log n > g = n(\log n)^{10}$$

$$n \times \cancel{n \log n} \qquad \cancel{n} \times \cancel{\log n} \times (\log n)^9$$

$$n > (\log n)^9$$

$9 > 8$

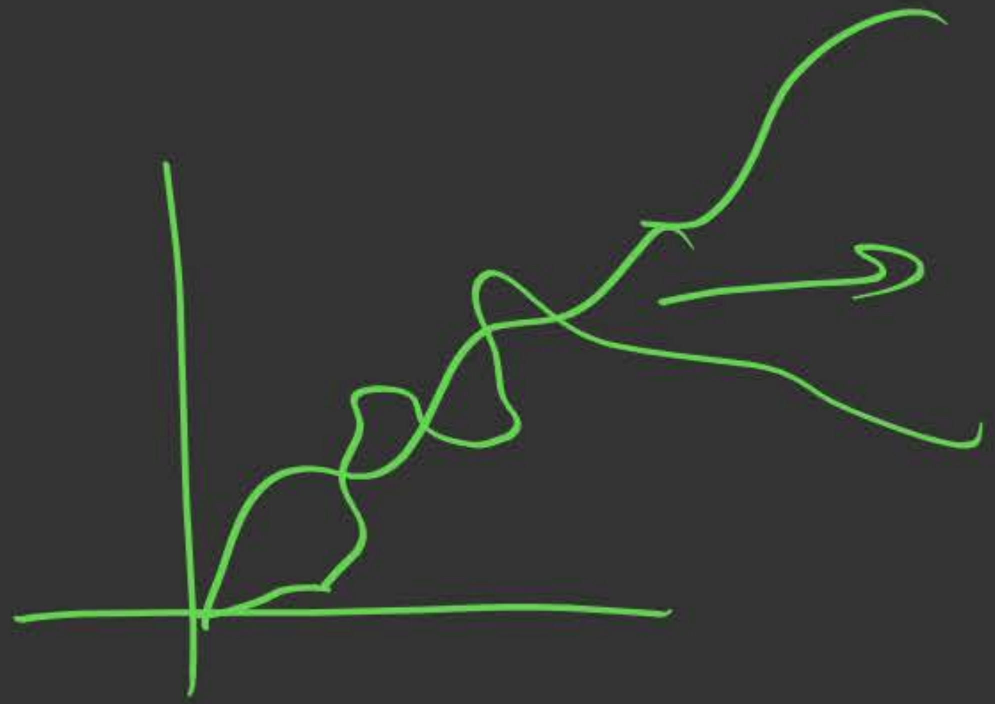
$n^e > 2^n$

$n = 1$

$n = 2$

$n = 3$

large n



[MSQ]

V. Interesting   V. Adv

#Q. Asymptotic Comparison of 2 functions:

$f(n) = n^3, 0 < n < 10{,}000$ 0 – 100

101 – 10000

$= n, n > 10{,}000$ > 10000

$g(n) = n, 0 < n \le 100$ Ans:- C, D

$= n^3, n > 100$ 0 – 100

101 – 10000

> 10000

A. $f(n) = \Omega(g(n))$, for $n > 100$

B. $f(n) = o(g(n))$, for $n > 100$

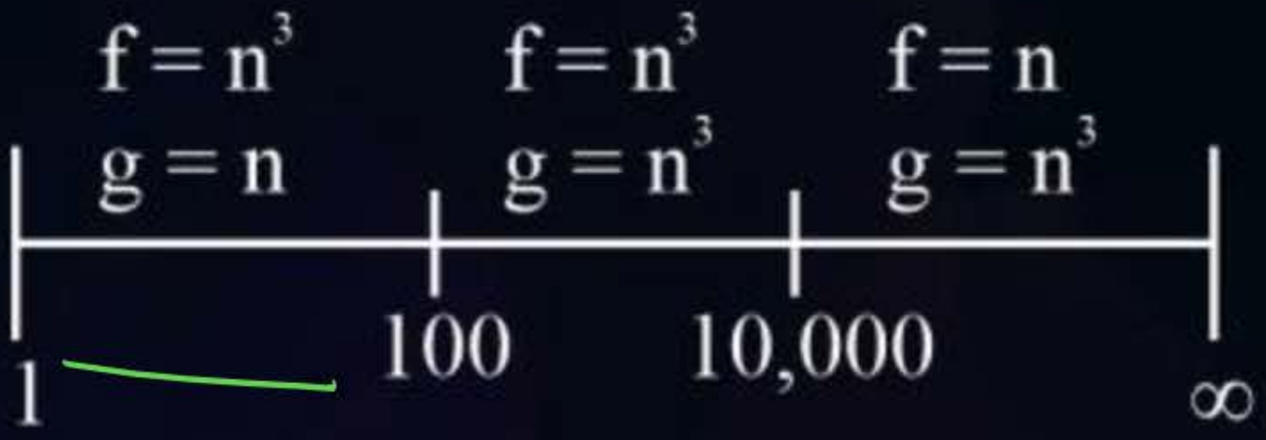
C. $f(n) = O(g(n))$, for $n > 100$

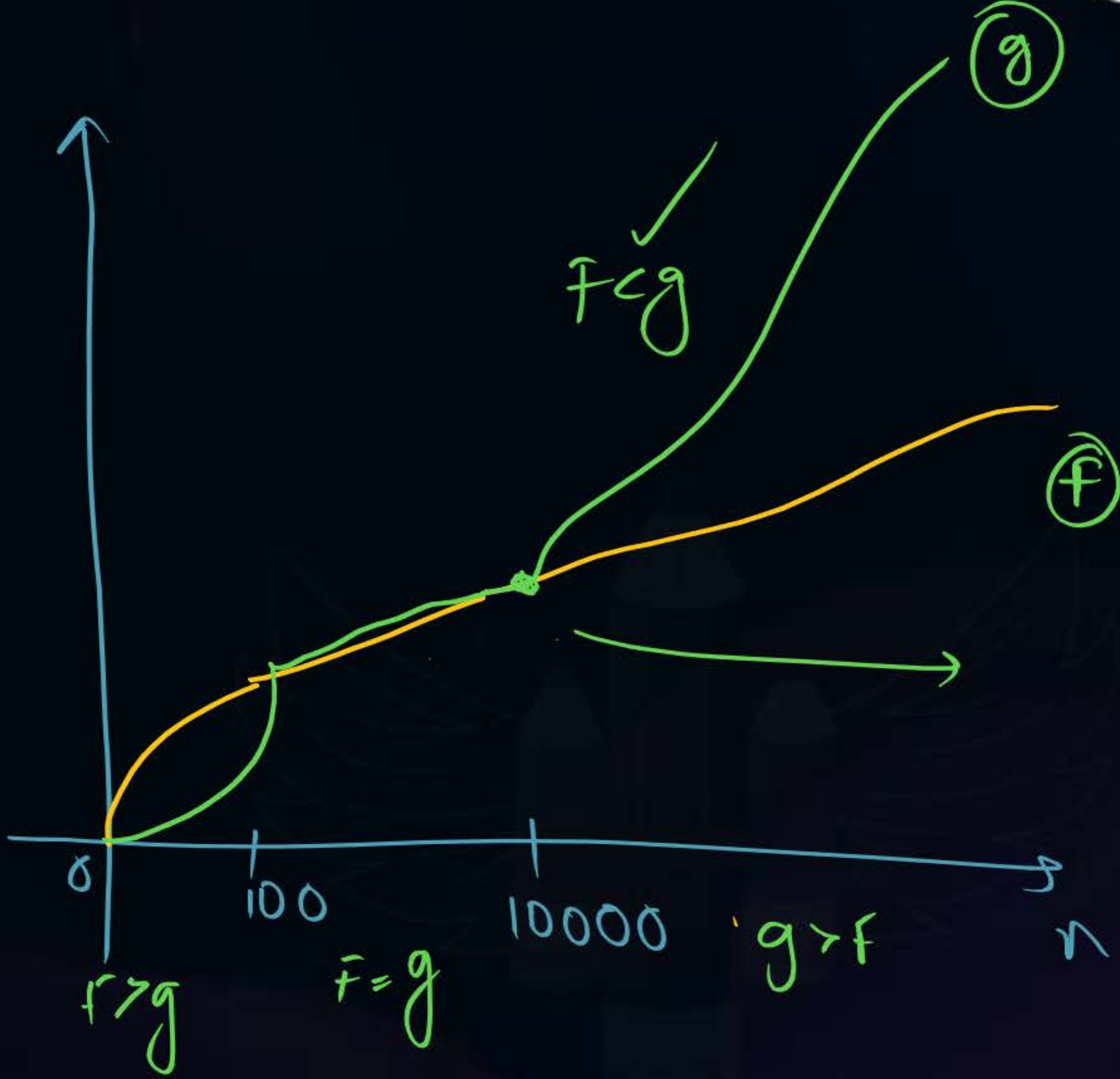
D. $f(n) = o(g(n))$, for $n > 10{,}000$

$f \geq g$

c) $f \geq g$, $\underline{n > 100}$

B) $f > g$, $n > 100$



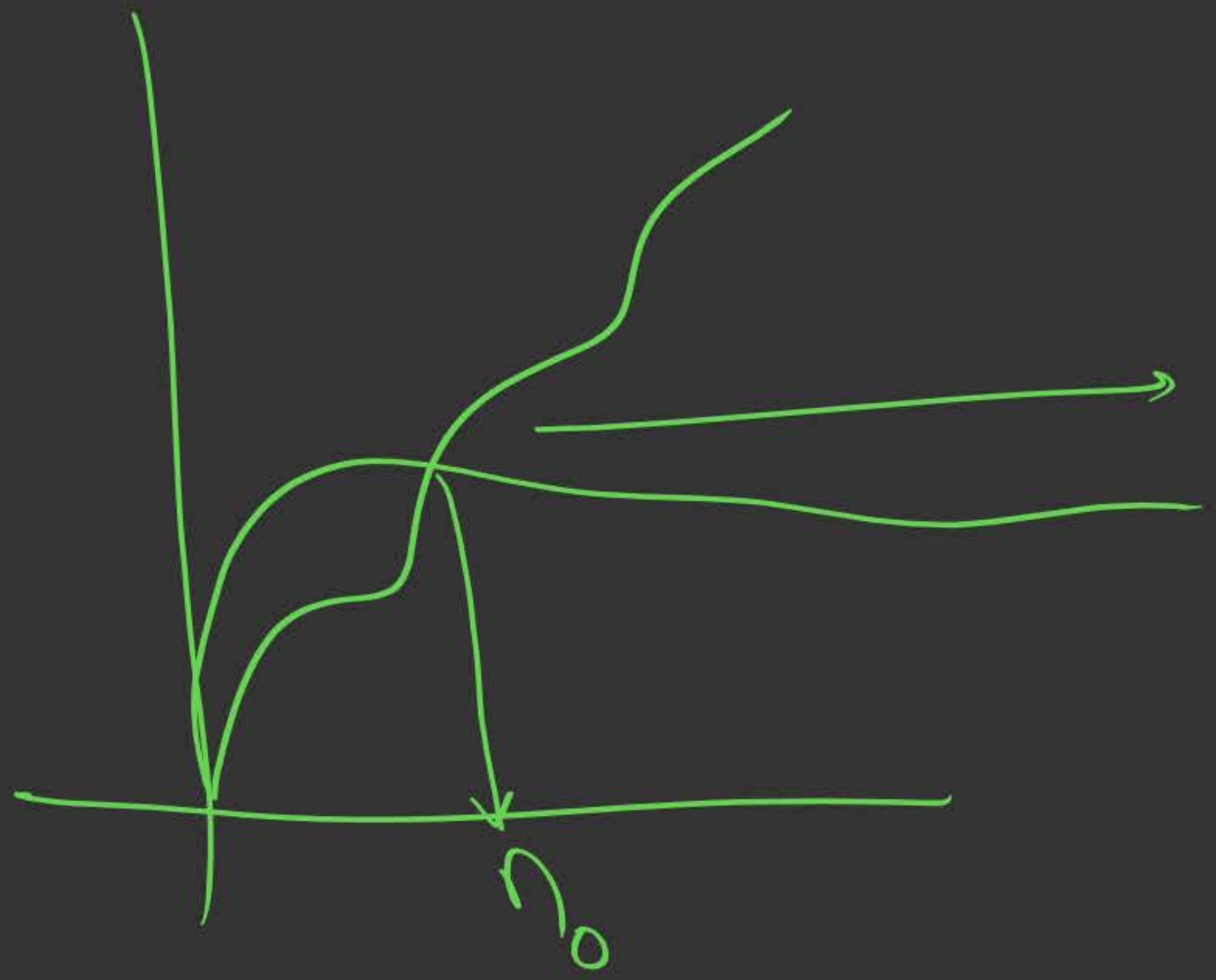


$$f = o(g)$$

iff $f < c * g$

$\boxed{n \geq n_0}$

threshold



[NAT]

Imp

#Q. You are given a database having 10^x records.

There are 2 packages available for processing the data.

Package A takes a time of $10 * n * \log n$ while,

Package B takes a time of $0.0001 * n^2$ for processing 'n' records.

Determine the smallest integer x for which package A outperforms package B.

22.7%

better

$t_A < t_B$

Ans: 6

$t_A = 10n\log n$

$t_B = 0.0001n^2$

$= 10^{-4}n^2$

$t_A < t_B$

$n = 10^x$

$t_A = 10 \times 10^x \times \log_{10}(10^x)$

$= 10^{x+1} * x$

$t_B = 10^{-4} \times (10^x)^2$

$= 10^{-4} \times 10^{2x}$

$= 10^{2x-4}$

$t_A < t_B$

$$\boxed{x \times 10^{x+1} < 10^{2x-4}}$$

$$x < \frac{10^{(2x-4)}}{10^{(x+1)}}$$

$$x < 10^{(2x-4-x-1)}$$

$$\boxed{x < 10^{(x-5)}}$$

Put $\underline{x = 5}$

$5 < 10^{5-5}$

$5 < 10^{0}$

$5 < 1$ ✗

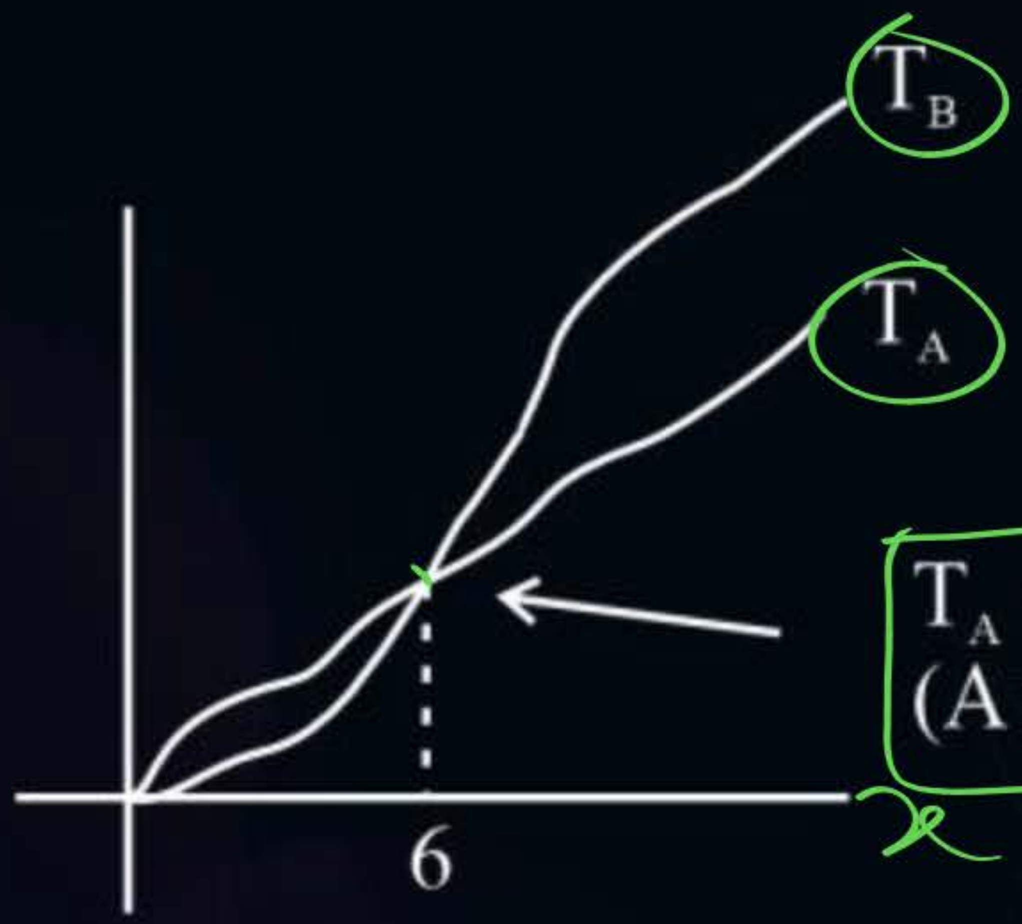
Put $\underline{x = 6}$

$6 < 10^{6-5}$

$6 < 10^{1}$

$6 < 10$ ✓

True



$T_A < T_B$
(A is better)

Approz:-

$x=1$	t_A	t_B
$x=2$	t_A	t_B
$x=3$		
$\vdots$		

[MCQ]

HW

#Q. Arrange the given functions in increasing order of rate of growth:

$f_1 \rightarrow 2^n, f_2 \rightarrow n^{3/2}, f_3 \rightarrow n \log n, f_4 \rightarrow n^{(\log n)}$

A f_2, f_3, f_1, f_4

B f_3, f_2, f_1, f_4

C f_3, f_2, f_4, f_1

D f_2, f_3, f_4, f_1

Summary

→ Practice Questions

→ PYQs

THANK - YOU